



Islamic University of Technology

Lab Report

Experiment No: 02

Student No: 210041105

Course Code: CHEM-4242

Name of the Experiment: STANDARDIZATION OF HCL
WITH STANDARD Na_2CO_3 SOLUTION

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Submitted by:

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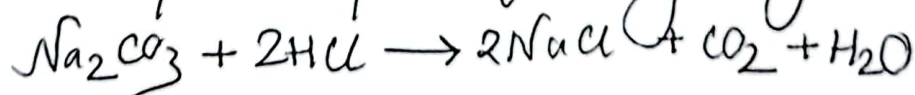
Objective:

To find the normality of an HCl solution using a standard solution of Na_2CO_3 solution.

Theory:

Double titration is an analytical chemistry technique that can be used to know the volume of an unknown acid or base. The first titration is done to standardize the titrant (usually taken in burette). The second titration is used to find the molarity of the titrand (unknown solution taken in a conical flask). Usually, methyl orange and phenolphthalein are used in double titration.

The reaction occurs in the following way:



Apparatus:

1. Conical flask
2. Pipette
3. Burette.
4. Funnel.
5. Stand and clamp

Chemicals used:

1. 0.53 g anhydrous Na_2CO_3 .
2. HCl solution
3. Phenolphthalein
4. Methyl orange.

Procedure:

A. Preparation of Na_2CO_3 soln:

We transfer 0.53 gm of anhydrous Na_2CO_3 in a 100 mL measuring flask and then dissolve it in distilled water and it is marked upto a certain level. The normality of the prepared solution is:

$$N_{\text{base}} = \frac{\text{weight taken (gm)} \times 0.1}{0.53}$$

B. Finding out concentration using titration method:

We took 10 mL of Na_2CO_3 in conical flask and it is diluted upto 50 mL. After that, one or two drops of phenolphthalein is added to that solution. It turns bright pink in colour. Then, it is titrated using the HCl solution in a burette. Addition of HCl is stopped when the pink colour vanishes. Using the first titration, we find the first end point. Again, 2-3 drops of methyl orange is added and titration is continued further using HCl solution. When the yellow colour of the solution turns to a faint reddish colour, we stop our titration to find the second end point. The difference between the end points determines the volume of HCl. This process is repeated thrice to find the mean reading. The strength of diluted

HCl was calculated finally. And is compared with the strength of supplied HCl solution.

Experimental Data:

No. of obs.	Volume of Na ₂ CO ₃ solution (ml)	Burette Reading (mL)			Difference = First End Point - Initial (mL)	Difference = Second End Point - Initial (mL)	Avg. of First Titration	Avg. of Second Titration
		Initial	First End Point	Second End Point				
1.	10.0	0.10	8.20	15.80	8.10	15.70	7.96	15.73.
2.		15.80	23.80	31.50	8.00	15.70		
3.		31.50	39.30	47.30	7.80	15.80		

$$\text{Initial weight} = 39.635 \text{ gm}$$

$$\text{Final weight} = 38.724 \text{ gm}$$

$$\text{Difference} = 0.911 \text{ gm}$$

$$\therefore \eta_{\text{et}} \text{ Concentration} = \frac{0.911}{0.53} \times 0.1$$

$$= 0.1718$$

Calculation:

Actual concentration of HCl = 8.77 N.

1.2 mL concentrated HCl diluted to 100 mL
dilute HCl (supplied)

$$\begin{aligned} V_{\text{dil. HCl}} \times S_{\text{dil. HCl}} &= V_{\text{Na}_2\text{CO}_3} \times S_{\text{Na}_2\text{CO}_3} \\ \Rightarrow S_{\text{dil. HCl}} &= \frac{V_{\text{Na}_2\text{CO}_3} \times S_{\text{Na}_2\text{CO}_3}}{V_{\text{dil. HCl}}} \\ &= \frac{10 \times 0.1718}{15.73} \\ &= 0.1079 \text{ N} \end{aligned}$$

$$\begin{aligned} V_{\text{conc. HCl}} \times S_{\text{conc. HCl}} &= V_{\text{dil. HCl}} \times S_{\text{dil. HCl}} \\ \Rightarrow S_{\text{conc. HCl}} &= \frac{V_{\text{dil. HCl}} \times S_{\text{dil. HCl}}}{V_{\text{conc. HCl}}} \\ &= \frac{0.1079 \times 100}{1.2} \\ &= 8.985 \text{ N} \end{aligned}$$

Results:

Concentration of HCl is 8.985 N

Error Calculation:

The known value of concentration = 8.77 N

The practical value obtained from experiment = 8.985 N.

$$\therefore \% \text{ error} = \frac{|\text{Known Value} - \text{Observed Value}|}{\text{Known Value}} \times 100\%$$

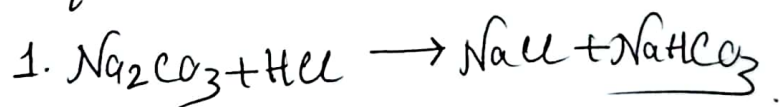
$$= \frac{|8.77 - 8.985|}{8.77} \times 100\%$$

$$= 2.1\%$$

Discussion:

All equipments must be thoroughly cleaned with distilled water before starting the experiment. While we realize that the endpoint is near, we need to carefully add the HCl drop by drop into the conical flask for both titrations.

When the colour of the indicators change, the titration needs to be stopped immediately. In the first titration, we used phenolphthalein as it has a pH range of 8.3-10. And it can detect in:



In the second titration, methyl orange was used as it has a pH range of 3-4.5 and can detect an acid like bicarbonate.

